Sols to ex. from Jan 19

Ex 1 $(x^4-1) = (x-1)(x+1)(x^2+1)$ irred in $\mathbb{Q}[x]$

$$\frac{1}{x^4-1} = \frac{1/2}{x^2-1} - \frac{1/2}{x^2+1} = \frac{1/4}{x-1} - \frac{1/4}{x+1} - \frac{1/2}{x^2+1}$$

$$\frac{x^5-4x^2+2}{x^4-1} = x + \frac{-4x^2+x+2}{x^4-1} = x + (-4x^2+x+2)\left[\frac{1/4}{x-1} - \frac{1/4}{x+1} - \frac{1/2}{x^2+1}\right] = x - \frac{4x-3}{4} - \frac{1/4}{x-1}$$

$$+ \frac{(-4x+5)}{-4} + \frac{3/4}{x+1} - \frac{4}{2} + \frac{(x+6)(-1/2)}{x^2+1} = x + \frac{-1/4}{x-1} + \frac{3/4}{x+1} + \frac{-\frac{x}{2}-3}{x^2+1}$$

Ex 2
$$R(f,g) = \begin{vmatrix} 1 & 0 & 0 & a & 0 \\ 0 & 1 & 0 & 0 & a \\ 1 & b & 1 & 0 & 0 \\ 0 & 1 & b & 1 & 0 \\ 0 & 0 & 1 & b & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & a \\ b & 1 & 0 & 0 \\ 1 & b & 1 & 0 \\ 0 & 1 & b & 1 \end{vmatrix} + \begin{vmatrix} 0 & 0 & a & 0 \\ 1 & 0 & 0 & a \\ 1 & b & 1 & 0 \\ 0 & 1 & b & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ b & 1 & 0 \\ 1 & b & 1 \end{vmatrix} - a\begin{vmatrix} b & 1 & 0 \\ 1 & b & 1 \\ 0 & 1 & b \end{vmatrix} + a\begin{vmatrix} 1 & 0 & a \\ 1 & b & 0 \\ 0 & 1 & 1 \end{vmatrix}$$

$$= 1 - a(b^3-2b) + a(b+a) = a^2 + a(-b^3+3b) + 1 \Rightarrow R(f,g) = 0 \text{ if}$$

$$a = \frac{(b^3-3b) \pm \sqrt{(b^3-3b)^2-4}}{2}$$

Ex 3 i) By Ex 17 III $\mathbb{Z}_p$ is local with max ideal $p\mathbb{Z}_p$. It is easy to see the $\mathbb{Z}_p$ is complete. By the last 2 lines of page 206, $\mathbb{Z}_p[[X]]$ is a complete local ring with maximal ideal (m, X).

ii) $k[X_1 \ldots X_n]_{(X_1 \ldots X_n)}$ is a local ring, but it is not complete

Think of $k[X_1 \ldots X_n]_{(X_1 \ldots X_n)}$ as a subring of $k[[X_1 \ldots X_n]]$

then e^x gives a Cauchy sequence in $k[x]$ hence in $k[x]_{(x)}$. So it suffices to show $e^x \notin k[x]_{(x)}$. But if $e^x = f(x)/g(x)$ then $g(x)\cdot e^x = f(x)$. But it is easy to see that $g(x)\cdot e^x$ can't be a non-zero polynomial.

iii) $K(X_1 \ldots X_n)$ is a field with maximal ideal $m=(0)$. So it's a local ring. Any Cauchy sequence is a constant sequence, so it's also complete